Final Exam – Summer 2025 Structure Breakdown

Q1 – Clinical Trial Power & Statistical Tests (15 pts) - Topic: Sample size, statistical power, t-test, chi-square for proportions, multiple comparisons. - Related lectures: * Sample Size and Power (PPTX) * Comparison of 2 Means (PPTX) * Proportions (PPTX #9 & #10) - Parts: 1. Interpret meaning of 90% power. 2. Effect of smaller SD on power. 3. Appropriate test for comparing means between 2 groups. 4. Appropriate test for comparing proportions across 4 groups. 5. Problem with multiple pairwise proportion tests.

Q2 – Odds Ratio, Relative Risk, Chi-Square (15 pts) - Topic: 2×2 tables, odds ratio vs relative risk, hypothesis testing, chi-square test. - Related lectures: * Proportions (PPTX #9 & #10) * Chi-square (found in Lesson 10 summary) - Parts: 1. Compute OR and RR. 2. Choose appropriate measure without knowing study design. 3. Null hypothesis formulation. 4. Compute expected counts under null. 5. Compute chi-square statistic.

Q3 – Logistic Regression & Survival Analysis (10 pts) - Topic: Logistic regression vs linear regression, censoring, time-to-event analysis. - Related lectures: * Logistic Regression (PPTX #13) * Survival Analysis (in summaries) - Parts: 1. Why linear regression is inappropriate; alternative method. 2. Assumption violated by censoring approach. 3. Test for comparing time-to-event.

Q4 – True/False Concept Check (20 pts) - Topic: Regression, hypothesis testing, Wilcoxon, diagnostic test properties, correlation, p-values, logistic regression, survival analysis, confidence intervals. - Related lectures: Multiple lectures 1–13 (comprehensive concepts). - Parts: 10 T/F statements.

Q5 – Simple Linear Regression (10 pts) - Topic: Regression interpretation, slope meaning, significance testing, predictions. - Related lectures: * Correlation and Regression (PPTX #11) - Parts: 1. Fitted equation. 2. Difference for 1-year age change. 3. Test significance of slope. 4. Difference for 10-year age change.

Q6 – Multiple Regression (15 pts) - Topic: Multiple regression interpretation, adjusted R^2, effect of covariates. - Related lectures: * Multiple Regression (PPTX #12) - Parts: 1. % variability explained. 2. Predicted difference for family history variable. 3. Significance of age after adjustment. 4. Compare effect of weight in simple vs multiple regression.

Q7 – Lassa Fever Pilot Study (15 pts) - Topic: Descriptive statistics, t-test, adjusted analysis, sample size estimation for power. - Related lectures: * Comparison of 2 Means (PPTX #6) * Sample Size and Power (PPTX) - Parts: 1. Summarize pilot data. 2. Compare CT values between groups; adjusted vs unadjusted. 3. Estimate SD, compute required n for 90% power to detect mean difference of 4.